LESSON 11.3
CONSTRUCTING SCATTER PLOTS

LESSON INTRODUCTION

MA.8.DP.1.1 Given a set of real-world bivariate numerical data, construct a scatter plot or a line graph as appropriate for the context.

MA.8.DP.1.2 Given a scatter plot within a real-world context, describe patterns of association.

LEARNING TARGETS

- I can describe patterns of association for the data using the words "increasing," "decreasing," "linear," or "non-linear."
- I can construct a scatter plot given a set of real-world data.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data, and on whether it is univariate or bivariate.

MA.912.DP.2.4 Fit a linear function to bivariate numerical data that suggests a linear association, and interpret the slope and y-intercept of the model. Use the model to solve real-world problems in terms of the context of the data.

ESSENTIAL QUESTIONS

- How do you create a scatter plot?
- How can a scatter plot show trends and associations?

LESSON NARRATIVE

Students build on their previous knowledge of scatter plots to begin constructing complete graphs. Students will revisit trends as they apply to both independent and dependent variables. Additionally, students will add to their vocabulary terms that describe associations, and then use this information to construct and describe scatter plots.

COMPONENT SUMMARY

11.3.1 WARM-UP *(~5 MIN.)*

Notice differences

11.3.2 EXPLORATION *(10 MIN.)*

Analyze scatter plots

11.3.3 BRING IT TOGETHER *(5-10 MIN.)*

Notice patterns of association

11.3.4 GUIDED PRACTICE *(15-20 MIN.)*

Construct scatter plots

11.3.5 WRAP-UP *(~5 MIN.)*

Reflect on learning

RESOURCES

GLOSSARY

Key Terms:

- Association

Additional Terms: N/A

MATERIALS

- Printouts of data tables (one per group) for 11.3.2 Activity
- (Optional) Poster-size graph paper
- (Optional) Plain, white paper

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may believe that there can be no outliers in order for the data to be represented by a trend.
- Students may inadvertently believe that some associations are "better" than others.

THINGS TO CONSIDER

- Review the idea that a trend explains a general direction but may not necessarily capture every single data point. Review outliers and their role in data sets.
- Compare graphs with strong and weak associations. Ask students to describe each graph. Remind students that data is neutral and that neither association is better than the other.

LITERACY AND LANGUAGE ROUTINES

Three Reads

The Bring It Together introduces mathematical vocabulary with which students may be unfamiliar. Encourage students to read each portion of the text multiple times and to use correct mathematical vocabulary when discussing each item with their partner or small group.

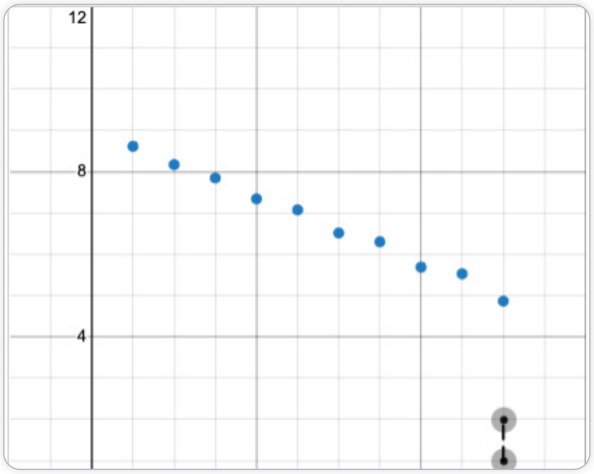
MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.1.1 Participate in Effortful Learning

This lesson develops students' ability to analyze and problem solve. Students must analyze problems in a way that makes sense given the task. Support students as they work to stay engaged and maintain a positive mindset as they work to solve the tasks.

DIFFERENTIATE INSTRUCTION

Consider providing Desmos Activity "Scatter Plot Capture" as support for remediating students' conceptual understanding of bivariate data. In this activity, students use observations about scatterplot relationships to make predictions about future points in the plot. In particular, students focus on linear vs nonlinear associations, strong vs weak associations, and increasing vs decreasing plots.



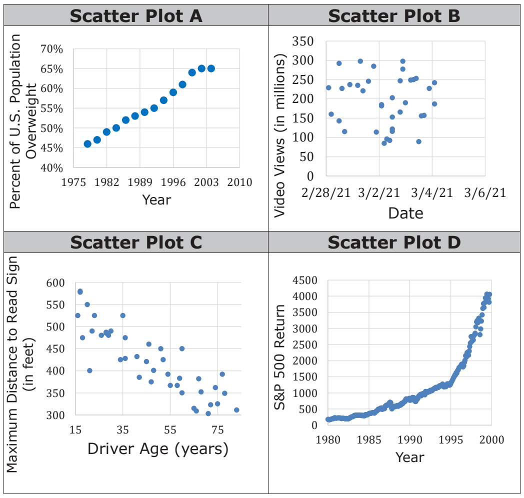
11.3.1 WARM-UP: WHICH ONE DOESN'T BELONG?

DIFFERENTIATE INSTRUCTION

Consider multiple options for completing the activity. Possibilities include students answering independently and then discussing as a whole group, students answering independently and then going to a particular corner of the room based on their answer to compare reasons for selecting that graph, and/or placing students in groups of four with each student assigned a graph to explain why it does not belong.

11.3.1 Warm-Up: Which One Doesn't Belong?

1. Four scatter plots are shown.



a. Which one doesn't belong?
Answers will vary.

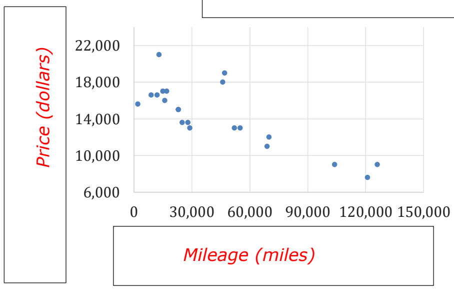
b. Justify your choice.
Scatterplot A: Only one close to linear.
Scatterplot B: Seems to be no correlation.
Scatterplot C: Only one clearly decreasing.
Scatterplot D: Only one that look exponential.

11.3.2 Exploration: Scatter Plots and Their Data

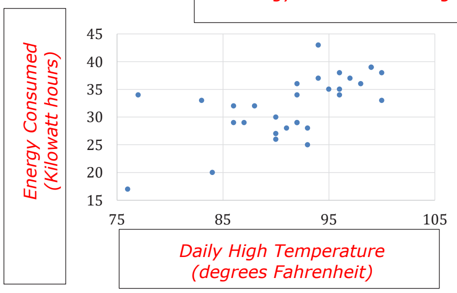
Your teacher will give you four tables of data.

1. Work with your partner to match each table with one of the scatter plots shown. Use the information from the tables to label the axes and title of each scatter plot.

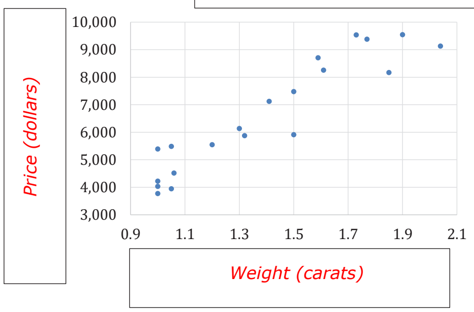
Scatter Plot I: *Used Cars*



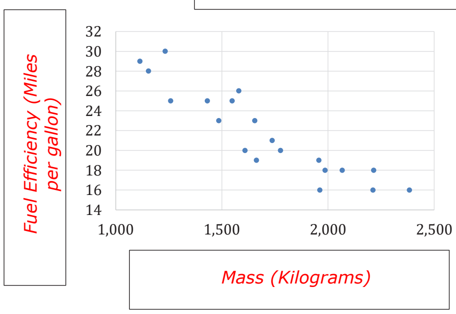
Scatter Plot II: *Energy Consumed in August*



Scatter Plot III: *Diamonds*



Scatter Plot IV: *New Cars*



11.3.2 EXPLORATION: SCATTER PLOTS AND THEIR DATA

TEACHER MOVES

Common Misconception

Students may believe that a trend line or line of fit must reflect every point on a graph. Have students draw a line of fit for Scatter Plot I. Discuss the trend of the data, but also ask students to notice points that seem to veer from the line of fit. Remind students that there may often be outliers to a trend and/or line of fit.

ENRICHMENT

Encourage students to analyze features of the graphs as they match the graphs to the labels:

- How would you describe Scatter Plot I?
- What is the independent variable? What is the dependent variable?
- How is the data on the y-axis dependent on the data on the x-axis? How would you describe the data on the scatter plot?

DIFFERENTIATE INSTRUCTION

Provide students with a word bank to use to label the graphs.

11.3.2 EXPLORATION: SCATTER PLOTS AND THEIR DATA (CONTINUED)

TEACHER MOVES

MA.K12.MTR.1.1: Participate in Effortful Learning

Foster students' ability to persevere when engaging in challenging tasks by encouraging students to ask questions that help them work through issues and solve a task at hand.

DIFFERENTIATE INSTRUCTION

If students struggle with flipping between pages, have one partner lay their graphs out while the other partner turns to the pages with the questions. Also, encourage students to verbally describe each graph before they begin answering the questions.

2. Work with your partner to complete each statement.

   a. The trend on scatter plot III shows that as the weight of a diamond increases, its price
      - ○ decreases.
      - ○ increases.

   b. The trend on scatter plot IV shows that as the mass of a new car increases, its fuel efficiency
      - ○ decreases.
      - ○ increases.

   c. The point $(76, 17)$ on scatter plot II represents that when the
      - ○ daily high temperature
      - ○ energy consumed for a home

      was 76
      - ○ degrees Fahrenheit (°F),
      - ○ kilowatt hours (kWh),

      the
      - ○ daily high temperature
      - ○ energy consumed for a home

      was 17
      - ○ °F.
      - ○ kWh.

11.3.3 Bring It Together: Scatter Plots and Their Data

Each point on a scatter plot represents a specific pair of values from the raw data. The x- and y-values are used to plot each data point in the coordinate plane. When all of the points in a data set are represented on a scatter plot, trends, or patterns of ***association***, can often be seen.

An ***association*** is a way to describe the form, direction, or strength of the relationship between the two variables in a bivariate data set. For numerical data, descriptions include linear or nonlinear; positive or negative; strong or weak.

1. Refer back to the scatter plots from the Warm-Up.

   a. Using your knowledge of linear and non-linear relationships, work with your partner to complete the table.

Scatter Plot(s) with a Linear Association	☐ Scatter Plot A ☐ Scatter Plot B ☐ Scatter Plot C ☐ Scatter Plot D
Scatter Plot(s) with a Non-Linear Association	☐ Scatter Plot A ☐ Scatter Plot B ☐ Scatter Plot C ☐ Scatter Plot D

Similar to the slope of a line, bivariate numerical data is said to have a positive association if as the x-values increase, the y-values also increase. If as the x-values increase, the y-values decrease, then the data is said to have a negative association.

   b. Work with your partner to complete the table for the scatter plots in the Warm-Up.

Scatter Plot(s) with a Positive Association	☐ Scatter Plot A ☐ Scatter Plot B ☐ Scatter Plot C ☐ Scatter Plot D
Scatter Plot(s) with a Negative Association	☐ Scatter Plot A ☐ Scatter Plot B ☐ Scatter Plot C ☐ Scatter Plot D
Scatter Plot(s) with an Association That Is Neither Positive nor Negative	☐ Scatter Plot A ☐ Scatter Plot B ☐ Scatter Plot C ☐ Scatter Plot D

If the points on the scatter plot closely follow a line or a curve, the association is described as strong. If the points on the scatter plot are more spread out, the association is described as weak.

   c. Work with your partner to complete the table for the scatter plots in the Warm-Up.

Scatter Plot(s) with a Strong Association	☐ Scatter Plot A ☐ Scatter Plot B ☐ Scatter Plot C ☐ Scatter Plot D
Scatter Plot(s) with a Weak Association	☐ Scatter Plot A ☐ Scatter Plot B ☐ Scatter Plot C ☐ Scatter Plot D

11.3.3 BRING IT TOGETHER: SCATTER PLOTS AND THEIR DATA

TEACHER MOVES

Three Reads

Encourage students to read each section of the text several times. Guide students as they make connections to other vocabulary and concepts, such as discussing other contexts in which they have heard the word association, and how those contexts are similar to this one.

ENRICHMENT

Encourage students to activate and apply prior knowledge through questioning:

- How do linear and non-linear relationships differ? How are they similar?
- How can a line of fit be used to explain and/or more clearly show relationships in a scatter plot?

TEACHER MOVES

Common Misconception

Students may incorrectly believe that positive associations are better than negative associations. Have students brainstorm data sets that show positive and negative associations. Have students describe why the data occurs as it does. Encourage students to see that positive or negative just explains the slope of the line and does not have any relationship to good or bad.

DIFFERENTIATE INSTRUCTION

Have students create a Frayer model to illustrate positive, negative, strong, and weak associations. Students will divide a piece of paper into fourths. In each section, students will write one of the terms, its definition, and draw an example of it. Also, encourage students to verbally explain how their illustrations demonstrate the concepts to a partner.

11.3.4 GUIDED PRACTICE: CONSTRUCTING SCATTER PLOTS

TEACHER MOVES

Students could argue either way. For scatter plots, there is not always a clear independent and dependent variable. In this case, it makes the most sense to show hits along the x-axis because as the number of hits increases, there is a greater likelihood (opportunity) for one of those hits to be a home run.

11.3.4 Guided Practice: Constructing Scatter Plots

1. The table shows the number of hits and home runs for professional baseball players on the Tampa Bay Rays who had at least 45 hits in the 2019 Major League Baseball (MLB) regular season.

Hits	Home Runs
154	33
80	17
107	19
155	21
82	14
138	20
86	16
135	20
102	14
55	3
46	5

a. What range of data values need to be represented along each axis to represent this data set?

Range of the Number of Hits	46 to 155
Range of the Number of Home Runs	3 to 33

b. Discuss with your partner which variable should be represented along each axis when constructing a scatter plot to represent this data.
Answers will vary.

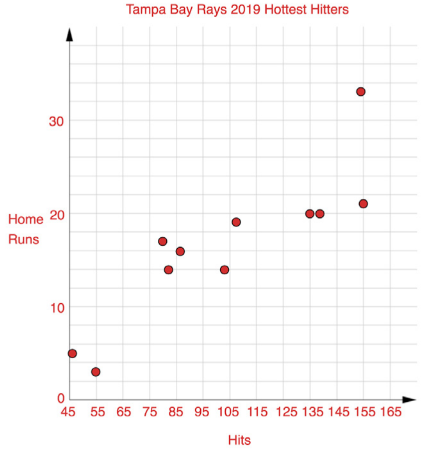
DIFFERENTIATE INSTRUCTION

Provide students with visual or fine motor challenges with poster-size graph paper to use.

TEACHER MOVES

Structurally, students may have used different scales on the graphs. If this occurs, encourage students to discuss the pros and cons of each scale. Also remind students to check that they graphed the points correctly.

c. Construct a scatter plot to represent the data by completing the following.
- Label the axes with the appropriate variables.
- Identify and label an appropriate scale for each axis on the grid.
- Plot a point for each pair of values in the data set.
- Title the graph.



d. Compare scatter plots with your partner. Discuss and resolve any differences, if necessary.
Answers will vary.

e. Work with your partner to describe the relationship between hits and home runs based on the data from the Tampa Bay Rays. Use at least two words from the word bank in your description.

decrease(s)	increase(s)	linear
non-linear	positive	negative

Answers will vary.
Sample response: There is a positive linear relationship between the number of hits and the number of home runs. In general, as the number of hits increases, the number of home runs increases as well.

f. Work with your partner to explain why a scatter plot better represents this data set than a line graph.
Answers will vary.

Sample Response: A scatter plot better represents the data than a line graph because it would not make sense to look from point to point. In this case, a scatter plot is a better representation because we want to see the overall trends for any player on the team.

g. Ask two classmates for the explanation they wrote. Record their explanations and write your summary of their reasons. *You are the only person who should write in the first two columns of the table.* Have the person initial next to your summary stating that your summary was correct.
Answers will vary.

11.3.4 GUIDED PRACTICE: CONSTRUCTING SCATTER PLOTS (CONTINUED)

TEACHER MOVES

Students may explore how a scatter plot shows trends, as opposed to how the line would appear in a line graph. Encourage students to discuss any differences in their responses.

11.3.4 GUIDED PRACTICE: CONSTRUCTING SCATTER PLOTS (CONTINUED)

TEACHER MOVES

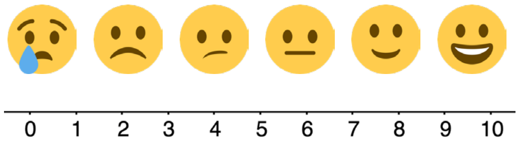
Remind students that an explanation tells why something occurs or why someone thinks as they do. Encourage students to explore the idea that they may make the same choices with their graphs but have different reasons or explanations for doing so.

Explanation	My Summary of Their Reason	Initials

h. Discuss the explanations from your classmates with your partner and if necessary, revise your explanation in Part F.
Answers will vary.

11.3.5 Wrap-Up: Reflection

1. Use the emoji scale to describe how confident you feel in constructing a scatter plot given a set of real-world data.
   Answers will vary.



2. Explain your selection.
   Answers will vary.

11.3.5 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

As an alternative to the Wrap-Up, consider placing a number line on a chart of paper for students to mark as they exit the classroom. This will provide a visual spread.